MODULE 1

Introduction to Renewable Energy: Overview of global energy demand and the need for renewable energy, Comparison of renewable and non-renewable energy sources, Environmental benefits and challenges of renewable energy.

Solar Radiation: Extra-terrestrial radiation, spectral distribution of extra-terrestrial radiation, solar constant, solar radiation at the earth's surface, beam, diffuse and global radiation.

Overview of global energy demand:

Global energy demand is steadily increasing, driven by population growth, economic development, and rising living standards. However, the current reliance on fossil fuels to meet this demand is unsustainable due to their finite nature, environmental impact, and contribution to climate change. This necessitates a transition towards renewable energy sources.

Renewable energy, derived from natural processes that are replenished constantly, offers a sustainable alternative to fossil fuels. Sources like solar, wind, hydro, geothermal, and biomass have the potential to meet a significant portion of global energy needs while mitigating environmental concerns.

The International Energy Agency (IEA) forecasts that global renewable electricity generation will increase by almost 90% from 2023 to 2030. This expansion is driven by factors such as falling costs of renewable technologies, supportive government policies, and growing public awareness of the need for clean energy.

Several milestones are expected to be reached in the coming years:

- In 2025, renewables-based electricity generation will overtake coal-fired generation.
- In 2026, wind and solar power generation will each surpass power generation from nuclear.
- In 2029, electricity generation from solar PV will surpass hydropower, becoming the largest renewable power source globally.

Despite this progress, challenges remain in fully transitioning to renewable energy. These include the intermittency of some renewable sources, the need for grid modernization, and the continued reliance on fossil fuels in certain sectors. However, with continued investment in research, development, and infrastructure, renewable energy can play a crucial role in meeting global energy demand sustainably.

Need for renewable energy:

The concern for the environment due to the ever-increasing use of fossil fuels and the rapid depletion of these resources has led to the development of alternative sources of energy, which are renewable and environmentally friendly. The following points may be mentioned in this connection.

1. The demand for energy is increasing by leaps and bounds due to rapid industrialization and population growth. Conventional sources of energy will not be sufficient to meet the growing demand.

2. Conventional sources (fossil fuels, nuclear) also cause pollution; thereby their use degrades the environment.

3. Conventional sources (except hydro) are non-renewable & bound to finish one day.

4. Large hydro-resources affect wildlife, cause deforestation & pose various social problems, due to the construction of big dams.

5. Fossil fuels are also used as raw materials in the chemical industry (for chemicals, medicines, etc.,) & need to be conserved for future generations.

Due to these reasons, it has become important to explore & develop non-conventional energy resources to reduce too much dependence on conventional resources. However, the present trend development of NCES indicates that these will serve as supplements rather than substitutes for conventional sources for some more time to time.

Salient features of non-conventional energy sources (NCES):

Merits:

1. NCES are available in nature, free of cost.

2. They cause no or very little pollution. Thus, by and large, they are environmentally friendly.

3. They are inexhaustible.

4. They have a low gestation period.

Demerits:

1) Though available freely in nature, the cost of harnessing energy from NCES is high, as in general, these are available in dilute forms of energy.

2) Uncertainty of availability: the energy flow depends on various natural phenomena beyond human control.

3) Difficulty in transporting this form of energy.

Advantages & disadvantages of conventional energy resources:

Advantages:

1) Coal: at present is cheap.

2) Security: by storing a certain quantity, the energy availability can be ensured for a certain period.

3) Convenience: it is very convenient to use.

Disadvantages:

1) Fossil fuels generate pollutants: CO, CO_2, NO_X, SO_X. Particulate matter & heat. The pollutants degrade the environment, pose health hazards & cause various other problems.

2) Coal: it is also a valuable petrochemical & used as the source of raw material for chemicals, pharmaceuticals & paints, industries, etc. From a long-term point of view, it is desirable to conserve coal for future needs.

3) Safety of nuclear plants: it is a controversial subject.

4) Hydro-electrical plants are the cleanest but large hydro reservoirs cause the following problems.

a) As large land area are submerged into water, it leads to deforestation.

b) Causes ecological disturbances such as earthquakes.

c) Causes dislocation of large populations & consequently their rehabilitation problems.

Comparison of renewable and non-renewable energy resources:

Feature	Renewable Energy	Non-Renewable Energy
Source	Naturally replenished resources (sunlight, wind, water, geothermal heat, biomass)	Finite resources that take millions of years to form (fossil fuels like coal, oil, natural gas; nuclear fuels like uranium)
Availability	Continuously replenished, virtually inexhaustible	Limited supply, depletes with use
Environmental Impact	Generally low environmental impact, minimal greenhouse gas emissions, less pollution	Significant environmental impact, major source of greenhouse gas emissions, air and water pollution, habitat destruction
Cost	High initial investment for infrastructure, but low operating costs and decreasing technology costs	Lower initial costs, but fluctuating fuel costs and potentially higher long-term costs due to environmental damage and resource scarcity
Reliability	Some sources (solar, wind) are intermittent, requiring energy storage or backup systems; others (hydro, geothermal) are more consistent	Generally reliable and dispatchable, but subject to fuel price fluctuations and potential supply disruptions
Land Use	Can require large land areas, but can also be integrated into existing infrastructure (e.g., rooftop solar)	Often requires less land area, but extraction and transportation can have

		significant environmental impacts
Job Creation	Growing sector with potential for significant job creation in manufacturing, installation, maintenance, and research	Mature sector with established infrastructure, but potential for job losses as the world transitions to renewable energy
Energy Security	Diversification of energy sources, reduced reliance on foreign fuel imports	Dependence on finite resources, geopolitical risks associated with fuel supply

Environmental Benefits and Challenges of Renewable Energy:

Renewable energy offers significant environmental benefits but also presents challenges that need to be addressed through technological advancements, sustainable resource management, and improved energy policies.

Environmental Benefits of Renewable Energy:

1. **Reduced Greenhouse Gas Emissions**
   - Renewable energy sources like solar, wind, hydro, and geothermal produce little to no CO_2 emissions, helping mitigate climate change.
2. **Lower Air Pollution**
   - Unlike fossil fuels, renewables do not release pollutants like sulfur dioxide (SO_2), nitrogen oxides (NO_x), or particulate matter, improving air quality and reducing respiratory diseases.
3. **Sustainable Resource Utilization**
   - Renewable sources are naturally replenished, reducing dependence on finite fossil fuel reserves.
4. **Reduced Water Consumption**
   - Wind and solar power require minimal water compared to thermal power plants (coal, nuclear), which use large amounts of water for cooling.
5. **Preservation of Ecosystems**
   - By reducing mining and drilling activities associated with fossil fuels, renewables help protect biodiversity and natural habitats.
6. **Energy Security and Decentralization**
   - Localized renewable energy production reduces reliance on imported fuels and enhances energy security.

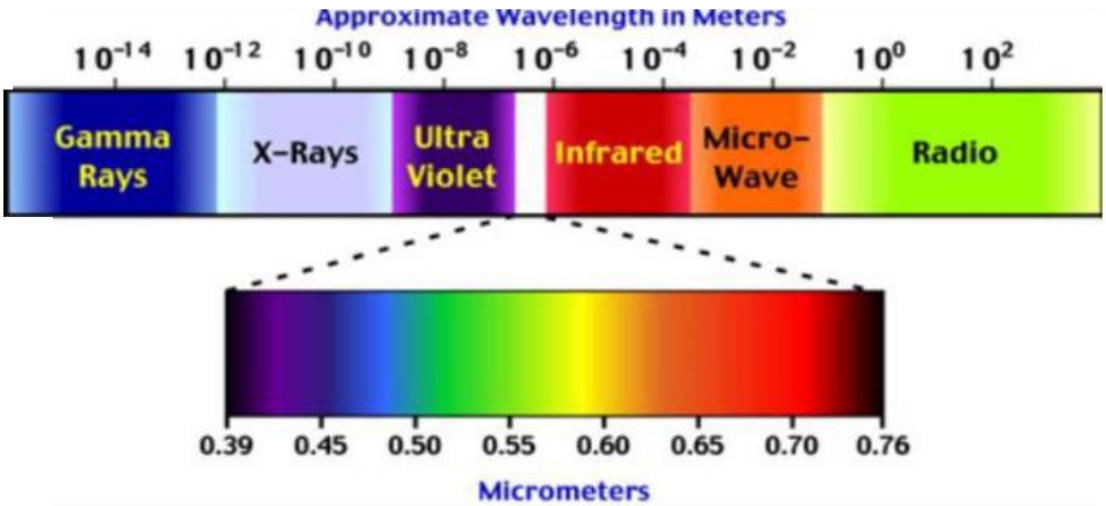
Environmental Challenges of Renewable Energy:

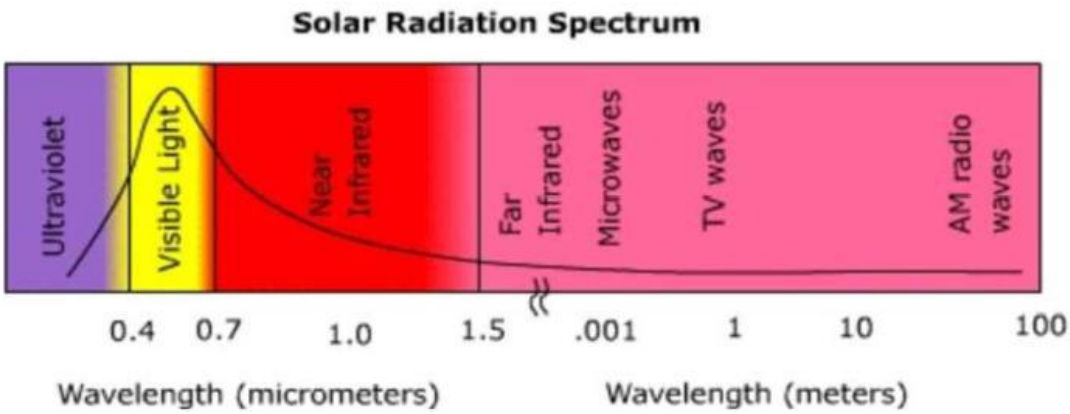
1. **Land Use and Habitat Disruption**
   - Large-scale solar farms and wind farms require significant land, potentially impacting wildlife and ecosystems.

2. **Intermittency and Reliability**
    - Solar and wind energy depend on weather conditions, requiring energy storage solutions or backup power sources.
3. **Resource Extraction for Technology**
    - Manufacturing solar panels, wind turbines, and batteries require rare earth metals and materials like lithium, which involve mining impacts.
4. **End-of-Life Waste Management**
    - Solar panels, wind turbine blades, and batteries pose disposal and recycling challenges at the end of their lifecycle.
5. **High Initial Costs**
    - The upfront investment for renewable infrastructure can be high, though costs are decreasing over time.
6. **Ecological Impact of Hydropower**
    - Large dams can disrupt river ecosystems, fish migration, and local communities.

Solar Radiation:

The sun radiates energy uniformly in all directions in the form of electromagnetic waves. When absorbed by a body, it increases its temperature. It provides the energy needed to sustain life in our solar system. It is a clean, inexhaustible, abundantly and universally available renewable energy source. Major drawbacks of solar energy are: that it is a dilute form of energy, which is available intermittently, uncertainly and not steadily and continuously. However, it is more predictable than wind energy. Also, peak solar insolation (incident solar radiation) often coincides with peak daytime demand; it can be well-matched to commercial





power needs. The output of the sun is 2.8×10^{23} kW. The energy reaching the earth is 1.5×10^{18} kWh/year.

Solar energy can be utilised directly in two ways: (i) by collecting the radiant heat and using it in a thermal system or (ii) by collecting and converting it directly to electrical energy using a Photovoltaic system. The former is referred to as 'Solar Thermal' and the later as 'Solar Photovoltaic' (SPV) system.

The transfer of energy via electromagnetic waves that travel at the speed of light. The velocity of light in a vacuum is approximately 3 x 108 m/s. The time it takes light from the sun to reach the Earth is 8 minutes and 20 seconds. Heat transfer by electromagnetic radiation can travel through empty space. Anybody above the temperature of absolute zero (-273.15 ^{0}C) radiates energy to their surrounding environment. The many different types of radiation are defined by their wavelength. Electromagnetic radiation can vary widely.

Solar constant (I_{SC}):

The **solar constant, I_{SC}** is defined as the energy received from the sun per unit time, on a unit area of surface perpendicular to the direction of propagation of the radiation at the top of the atmosphere and at the earth's mean distance from the sun. The World Radiation Center (WRC) has adopted the value of the solar constant as 1367 W/m^2. This has been accepted universally as a standard value of solar constant.

Extraterrestrial and Terrestrial Radiation:

Solar radiation incident on the outer atmosphere of the earth is known as **extra-terrestrial radiation**. The extraterrestrial radiation deviates from the solar-constant value due to two reasons. The first is the variation in the radiation emitted by the sun itself. The variation due to

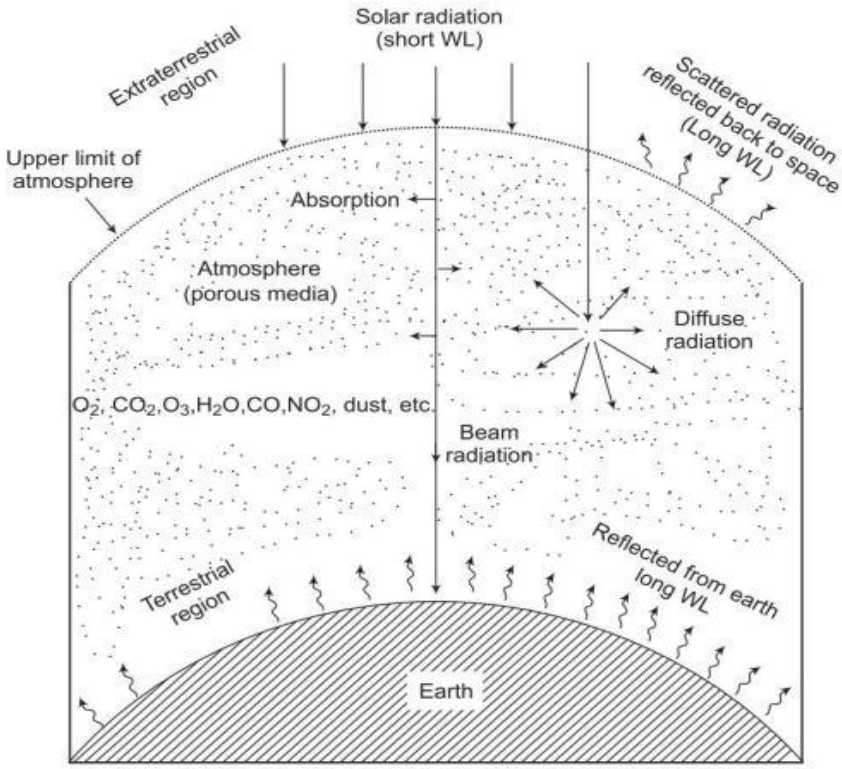


Figure Propagation of solar radiation through atmosphere

this reason is less than 1.5% with different periodicities. The second is the variation of the earth-sun distance arising from the earth's slightly elliptic path. The variation due to this reason is + 3%.

The extraterrestrial radiation, being outside the atmosphere, is not affected by changes in atmospheric conditions. While passing through the atmosphere it is subjected to mechanisms of atmospheric absorption and scattering depending on atmospheric conditions, depleting its intensity. A fraction of scattered radiation is reflected back to space while the remaining is directed downwards. Solar radiation that reaches the earth's surface after passing through the earth's atmosphere is known as **Terrestrial Radiation**. The terrestrial radiation expressed as energy per unit time per unit area (i.e. W/m^2) is known as Solar Irradiation. The term Solar insulation (incident solar radiation) is defined as solar radiation energy received on a given surface area in a given time (in J/m^2 or kWh/m^2). The positions of extraterrestrial and terrestrial regions are indicated in the below figure.

Spectral distribution of extra-terrestrial radiation:

Solar radiation covers a continuous spectrum of electromagnetic radiation in a wide frequency range. About 99 % of the extraterrestrial radiation has wavelengths in the range from 0.2 to 4 μm with maximum spectral intensity at 0.48 μm (green portion of visible range). About 6.4 % of extraterrestrial radiation energy is contained in the ultraviolet region (l < 0.38 μm); another 48 %t is contained in the visible region (0.38 μm < l < 0.78 μm) and the remaining 45.6 % is contained in the infrared region (l > 0.78 μm). There is almost complete absorption of short wave radiation in the range (l < 0.29 μm) and infrared radiation in the range (l > 2.3 μm) in the atmosphere. Thus, from the point of view of terrestrial applications of solar energy, the radiation only in the range of wavelengths between 0.29 and 2.3 μm is significant. The spectral solar irradiation distribution both for extraterrestrial and terrestrial radiation is shown in the below figure. The areas under these curves indicate the total radiation intensities in W/m^2 respectively for extraterrestrial and terrestrial regions.

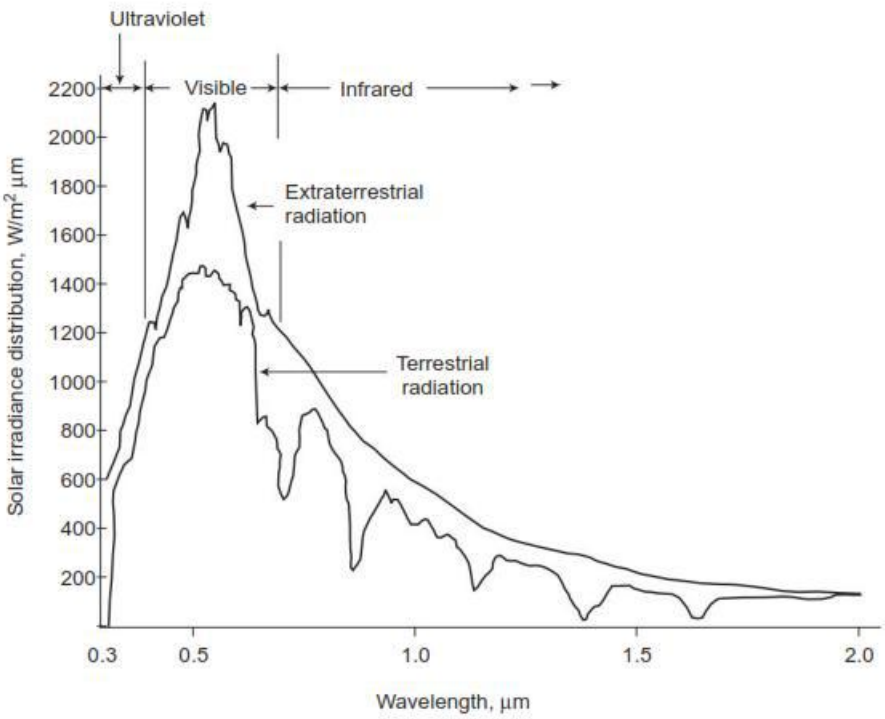
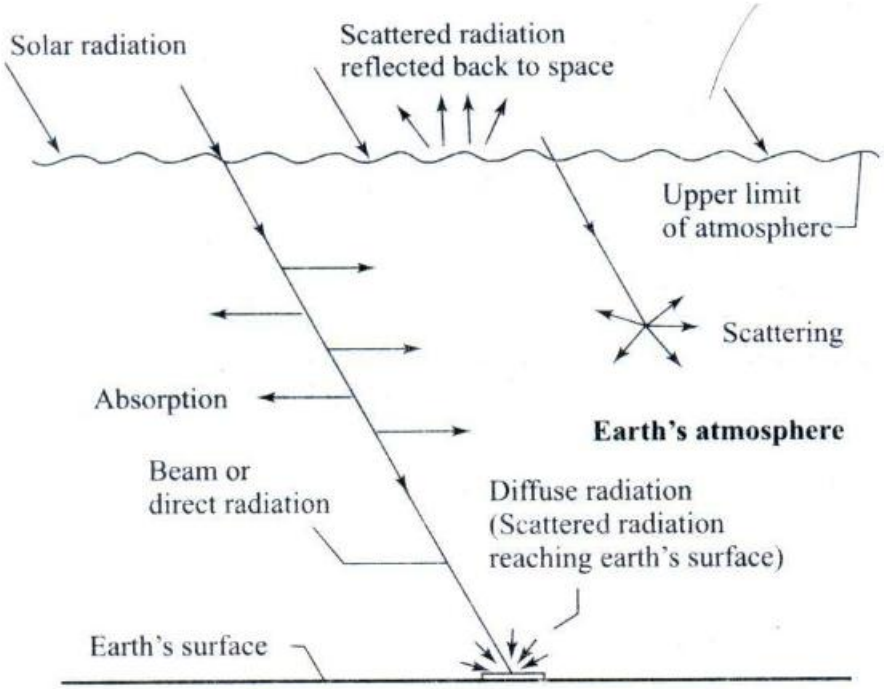


Figure Special solar irradiation, extraterrestrial and terrestrials

Solar radiation at the earth's surface:

Solar radiation received at the earth's surface is in the attenuated form because it is subjected to the mechanisms of absorption and scattering as it passes through the earth's atmosphere (Figure below). Absorption occurs primarily because of the presence of ozone and water vapor



in the atmosphere and a lesser extent due to other gases (like CO_2, NO_2, CO, O_2, and CH_4) and particulate matter. It results in an increase in the internal energy of the atmosphere. On the other hand, scattering occurs due to all gaseous molecules as well as particulate matter in the atmosphere. The scattered radiation is redistributed in all directions, some going back to space and some reaching the earth's surface. Solar radiation received at the earth's surface without change of direction i.e., in line with the sun is called direct radiation or beam radiation. The radiation received at the earth's surface from all parts of sky's hemisphere (after being subjected to scattering in the atmosphere) is called diffuse radiation. The sum of beam radiation and diffuse radiation is called as total or global radiation.

Beam, diffuse and global radiation:

Beam radiation: Solar radiation propagating in a straight line and received at the earth's surface without a change of direction, i.e., in line with the sun is called beam or direct radiation.

Diffuse radiation: Solar radiation scattered by aerosols, dust and molecules is known as diffuse radiation. It does not have a unique direction.

Global radiation: The sum of beam and diffuse radiation is referred to as total or global radiation.